



NOISE HARSHNESS VIBRATION

NHV Performance Analysis

Noise, Harshness & Vibration across the fleet, 2008–2018

Navigation

NHV Overview

Vibration Analysis

Noise and
Harshness
Analysis

Across the fleet **30.9%** of ride tests came back rough.

Average in-cabin noise sits at **60.6 dB**

Average customer rating is at **4.29** out of 5



NVH Overview

Year

All

Vehicle Type

All

Avg Cabin Noise

60.6 dB

Avg Vibration

0.85 m/s²

Avg Comfort Rating

4.29

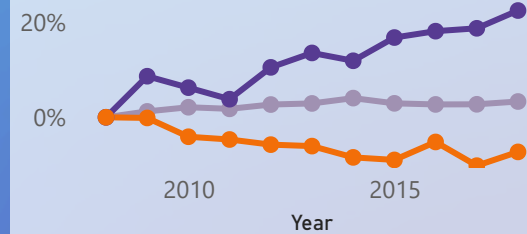
Critical Reviews

8.06%

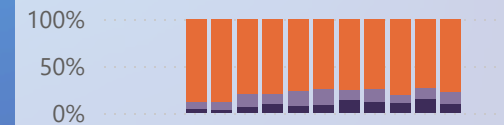
NHV Performance (2008-2018)

In reference with the year 2008 as Baseline

% Change ● Noise ● Vibration ● Comfort



Comfort Ratings



Noise Level by Component at different Speeds

● Engine/Powertrain ● Tire/Road ● Wind/Aero





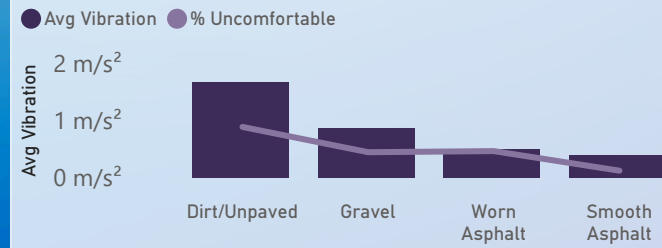
Vibration Performance and Statistics

Operating Condition Road Surface Type

All

All

Road Surface effect on Vibration



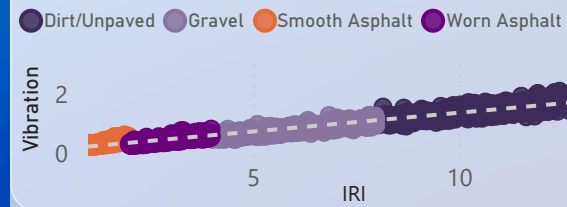
Avg Vibration

Vibration YoY%

0.85 m/s²

-1.26%

Vibration based on IRI and Road Surface



Mean Vibration

Std Vibration

0.63 m/s²

0.54 m/s²



Noise and Harshness Analysis

Year

All

Vehicle Type

All

Noise Levels by Speed and Component

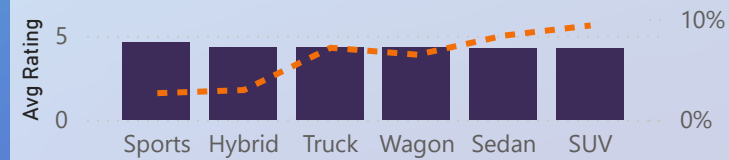
● Engine/Powertrain ● Tire/Road ● Wind/Aero



Model	Avg dB	Avg Rating	% Critical
4Runner SR5 V6	57.0 dB	4.57	6.90%
A4 3.0 4dr	59.5 dB	4.54	4.35%
Accord LX 2dr	59.5 dB	4.24	1.25%
Altima S 4dr	57.1 dB	4.41	4.55%
Total	60.6 dB	4.26	8.77%

Ratings by Vehicle Type

● Avg Rating ● % Critical Reviews



Average dB

60.6 dB

dB YoY%

-5.2%

Mean dB

63.3 dB

Std dB

9.4 dB



Analysis Insights and Recommendations

Every NHV metric improved from 2008 to 2018, noise down **3.3%** vibration down **22.5%** yet comfort ratings fell **-7.3%** and critical reviews more than doubled.

Road surface explains almost all the vibration variance. This prompts for a recommendation of benchmarking suspension tuning for vibration against matched road conditions since surface type alone explains most of the vibration variance.

SUVs are the quietest vehicles in the fleet and the most complained about. Sports cars are the loudest and the best loved. This clearly drives home a point that Noise in itself is not the issue. Vehicle manufactures should strive towards tuning sound to what each segment expects rather than just turning it down.

More Noise and Vibration reduction will not move the satisfaction number much further down the line. Perhaps, closing the gap between what the vehicle sensors read and what people actually feel in the seat will.

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